

$$\left\{ \frac{f(x) - f(a)}{x - a} \right\} \leftarrow \begin{array}{l} \text{absolute change} \\ \text{of } f. \end{array}$$

→ relative change in value of f

$$f'(a) := \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

$$= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$g'(a) = \dots \lim_{x \rightarrow a} [k(x+a)]$$

$$= k(a+a) = 2ka.$$

Warning:

It is not true that if $f = gh$.

then $f'(a) = g'(a)h'(a)$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$= \lim_{x \rightarrow a} \frac{(gh)(x) - (gh)(a)}{x - a}$$

$$= \lim_{x \rightarrow a} \frac{g(x)h(x) - g(a)h(a)}{x - a}$$

$$= \lim_{x \rightarrow a} \left[g(x) \frac{h(x) - h(a)}{x - a} + h(a) \frac{g(x) - g(a)}{x - a} \right]$$

$$= \lim_{x \rightarrow a} \left[g(x) \frac{h(x) - h(a)}{x - a} \right] + \lim_{x \rightarrow a} \left[h(a) \frac{g(x) - g(a)}{x - a} \right]$$

